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PUBLICATION

00-README

USAREUR-AF

Maven Smart System (MSS) — Training Curriculum

USAREUR-AF Operational Data Team

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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MAVEN SMART SYSTEM (MSS) — TRAINING CURRICULUM

USAREUR-AF OPERATIONAL DATA TEAM

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HEADQUARTERS, UNITED STATES ARMY EUROPE AND AFRICA
Wiesbaden, Germany
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Distribution: Distribution authorized to U.S. Government agencies and their contractors only. Other requests must be referred to Headquarters, USAREUR-AF, G6/Data, Wiesbaden, Germany.

New to MSS? Read **QUICK_START.md** first — operational in 30 minutes.

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HOW TO USE THIS CURRICULUM

Step 1 — Know Your Level

Level	Who You Are	Start Here
All personnel	Soldier, officer, or civilian accessing MSS for data consumption	TM-10
All staff (light builders)	Building dashboards, forms, or basic pipelines	TM-20 (prereq: TM-10)
Data-adjacent specialists	17/25-series, S6/G6, G2/G9, operational data analysts	TM-30 (prereq: TM-20)
WFF functional staff	Assigned to INT, FIRES, M2, SUST, PROTECTION, or MC	TM-40A–F (prereq: TM-30)
Technical specialists	Writing code, building external apps, ML models	TM-40G–L (prereq: TM-30)
Senior leaders (O-5+, CSM+)	Commanding or directing a data-capable formation	Data Literacy for Senior Leaders
All — background reading	Pre-MSS data concepts foundation	Data Literacy Technical Reference

TM-40 Disambiguation: Two distinct sub-series share the TM-40 designation. - **TM-40A–F** — Warfighting Function tracks. Prereq: TM-30. No coding required. Audience: functional WFF staff. - **TM-40G–L** — Technical Specialist tracks. Prereq: TM-30. Audience: engineers, analysts, and data professionals.

Confirm your sub-series against your role before enrolling.

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Step 2 — Follow the Learning Path

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Senior Leaders ──────────────────────────► Data Literacy for Senior Leaders
All Personnel ───────────────────────────► Data Literacy Technical Reference
[recommended]
All Personnel ───────────────────────────► TM-10 (Maven User)
                                                 │
                                                 ▼
All Staff ───────────────────────────────► TM-20 (Builder)
                                                 │
                                                 ▼
All proceeding to TM-40 ─────────────────► TM-30 (Advanced Builder)
                                            │            │
                                   ┌────────┘            └──────────┐
                                   ▼                               ▼
                    WFF Staff (TM-40A–F)              Technical Specialists (TM-40G–L)
                    TM-40A  Intelligence               TM-40G  ORSA
                    TM-40B  Fires                      TM-40H  AI Engineer
                    TM-40C  Movement/Maneuver          TM-40I  ML Engineer
                    TM-40D  Sustainment                TM-40J  Program Manager
                    TM-40E  Protection                 TM-40K  Knowledge Manager
                    TM-40F  Mission Command            TM-40L  Software Engineer
                                                            │
                                                            ▼
                                                   TM-50G–50L (Advanced)
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Step 3 — Reference the Glossary

Consult the Data & Foundry Glossary to translate general data concepts into MSS/Foundry terminology.

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PUBLICATIONS INDEX

Doctrine Publications

Publication	File	Audience
Data Literacy for Senior Leaders	doctrine/DATA_LITERACY_senior_leaders.md	O-5+, CSM+, Senior Civilians
Data Literacy Technical Reference	doctrine/DATA_LITERACY_technical_reference.md	All personnel
Glossary — Data & Foundry Terms	doctrine/GLOSSARY_data_foundry.md	All personnel
CDA Constraints and Directives	doctrine/CDA_CONSTRAINTS_AND_DIRECTIVES.md	TM-30+, all specialists

CDA Doctrine and Enterprise Architecture (TM-30, TM-40G–L)

Reference material supporting TM-30 and specialist tracks. Primarily targeted at TM-40G–L; WFF tracks (TM-40A–F) may use as supplemental reading.

Enterprise Architecture Series:

Publication	File	Supports
EA Reference Card	doctrine/enterprise_architecture/EA_00_REFERENCE_CARD.md	TM-30, TM-40K, TM-40L
EA Foundation	doctrine/enterprise_architecture/EA_01_FOUNDATION.md	TM-30
EA Schools of Thought	doctrine/enterprise_architecture/EA_02_SCHOOLS_OF_THOUGHT.md	TM-30, TM-40K
EA Artifacts and Views	doctrine/enterprise_architecture/EA_03_ARTIFACTS_AND_VIEWS.md	TM-40K, TM-40L
EA Governance	doctrine/enterprise_architecture/EA_04_GOVERNANCE.md	TM-40K
EA Military Application	doctrine/enterprise_architecture/EA_05_MILITARY_APPLICATION.md	TM-30, TM-40G–L

CDA Doctrine Series:

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Publication	File	Supports
Doctrine-Driven Development Overview	doctrine/cda_doctrine/CDA_DOCTRINE_OVERVIEW.md	TM-40G–L
CDA Doctrine Agent	doctrine/cda_doctrine/CDA_DOCTRINE_AGENT.md	TM-40H, TM-40L
AVT25 Assessment Case Study	doctrine/cda_doctrine/CDA_AVT25_ASSESSMENT.md	TM-40G, TM-40H
Identity vs. Classification	doctrine/cda_doctrine/CDA_IDENTITY_VS_CLASSIFICATION.md	TM-30, TM-40K, TM-40L

Technical Manuals — Foundational

Publication	File	Audience	Prereq
TM-10 — Maven User	tm/TM_10_maven_user/TM_10_MAVEN_USER.md	All staff	None
TM-20 — Builder	tm/TM_20_builder/TM_20_BUILDER.md	All staff	TM-10
TM-30 — Advanced Builder	tm/TM_30_advanced_builder/TM_30_ADVANCED_BUILDER.md	Data-adjacent specialists	TM-10, TM-20

TM-40A–F — Warfighting Function Tracks (Prereq: TM-30)

Publication	File	Concepts Guide	WFF
TM-40A — Intelligence	tm/TM_40A_intelligence/TM_40A_INTELLIGENCE.md	CONCEPTS_GUIDE_TM40A	INT
TM-40B — Fires	tm/TM_40B_fires/TM_40B_FIRES.md	CONCEPTS_GUIDE_TM40B	Fires
TM-40C — Movement & Maneuver	tm/TM_40C_movement_maneuver/TM_40C_MOVEMENT_MANEUVER.md	CONCEPTS_GUIDE_TM40C	M2
TM-40D — Sustainment	tm/TM_40D_sustainment/TM_40D_SUSTAINMENT.md	CONCEPTS_GUIDE_TM40D	Sustainment
TM-40E — Protection	tm/TM_40E_protection/TM_40E_PROTECTION.md	CONCEPTS_GUIDE_TM40E	Protection
TM-40F — Mission Command	tm/TM_40F_mission_command/TM_40F_MISSION_COMMAND.md	CONCEPTS_GUIDE_TM40F	MC

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TM-40G–L — Technical Specialist Tracks (Prereq: TM-30)

Publication	File	Concepts Guide	Audience
TM-40G — ORSA	tm/TM_40G_orsa/TM_40G_ORSA.md	CONCEPTS_GUIDE_TM40G	ORSA analysts
TM-40H — AI Engineer	tm/TM_40H_ai_engineer/TM_40H_AI_ENGINEER.md	CONCEPTS_GUIDE_TM40H	AI/ML specialists
TM-40I — ML Engineer	tm/TM_40I_ml_engineer/TM_40I_ML_ENGINEER.md	CONCEPTS_GUIDE_TM40I	MLEs
TM-40J — Program Manager	tm/TM_40J_program_manager/TM_40J_PROGRAM_MANAGER.md	CONCEPTS_GUIDE_TM40J	PMs, G8/S8
TM-40K — Knowledge Manager	tm/TM_40K_knowledge_manager/TM_40K_KNOWLEDGE_MANAGER.md	CONCEPTS_GUIDE_TM40K	KMOs, 37F
TM-40L — Software Engineer	tm/TM_40L_software_engineer/TM_40L_SOFTWARE_ENGINEER.md	CONCEPTS_GUIDE_TM40L	SWEs

TM-50G–L — Advanced Technical Specialist Tracks

Each TM-50 track builds on its TM-40G–L counterpart. Prereq: corresponding TM-40G–L.

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Publication	File	Concepts Guide	Audience	Prereq
TM-50G — Advanced ORSA	tm/TM_50G_orsa_advanced/TM_50G_ORSA_ADVANCED.md	CONCEPTS_GUIDE_TM50G	Senior ORSA	TM-40G
TM-50H — Advanced AI Engineer	tm/TM_50H_ai_engineer_advanced/TM_50H_AI_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50H	Senior AI engineers	TM-40H
TM-50I — Advanced MLE	tm/TM_50I_ml_engineer_advanced/TM_50I_ML_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50I	Senior ML engineers	TM-40I
TM-50J — Advanced PM	tm/TM_50J_program_manager_advanced/TM_50J_PROGRAM_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50J	Senior tech PMs	TM-40J
TM-50K — Advanced KM	tm/TM_50K_knowledge_manager_advanced/TM_50K_KNOWLEDGE_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50K	Senior KMOs	TM-40K
TM-50L — Advanced SWE	tm/TM_50L_software_engineer_advanced/TM_50L_SOFTWARE_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50L	Senior SWEs	TM-40L

TRAINING LEVEL DESCRIPTIONS

TM-10 — Maven User

Audience: All personnel. **Prereq:** None.

Competencies: CAC-based login; platform navigation; Workshop application use; authorized Action execution; Contour/Quiver no-code analysis; AIP/Agent interfaces; classification handling; access troubleshooting.

TM-20 — Builder

Audience: All staff. **Prereq:** TM-10.

Competencies: Foundry project creation via Compass UI; Pipeline Builder visual ETL; Ontology Manager Object Types and Link Types via UI; Action Editor; Workshop application build and publish; project permissions management; Foundry branching via UI; USAREUR-AF naming conventions.

TM-30 — Advanced Builder

Audience: Data-adjacent specialists (17/25-series, S6/G6, G2/G9). **Prereq:** TM-10, TM-20.

Competencies: Multi-page Workshop with conditional logic and variable passing; advanced Pipeline Builder multi-source joins and aggregations; Ontology design (Object Types, Link Types, Actions) via UI; advanced Contour/Quiver; AIP Logic workflow configuration (not authoring); data lineage review; governance workflows; branching and production promotion; C2DAO governance standards.

TM-40A–F — Warfighting Function Tracks

Prereq: TM-30. No coding required.

Track	WFF	Audience	Focus
TM-40A	Intelligence	G2/S2 staff, targeting officers, all-source analysts	Collection management, targeting, all-source analysis workflows
TM-40B	Fires	FSE, fire support officers, targeting teams	Targeting data, effects assessment, fire support coordination
TM-40C	Movement & Maneuver	G3/S3 maneuver staff	Maneuver tracking, route analysis, operational visualization
TM-40D	Sustainment	G4/S4, logistics staff	Logistics visibility, supply chain analytics, readiness tracking
TM-40E	Protection	Force protection, CBRN staff	Threat tracking, vulnerability assessment, protection COA support
TM-40F	Mission Command	C2/MC staff, battle captains	Operational dashboards, CCIR management, battle rhythm products

TM-40G–L — Technical Specialist Tracks

Prereq: TM-30.

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Track	Audience	Key Competencies
TM-40G ORSA	FA49, quantitative analysts	Code Workspaces (Python/R); regression, classification, time series, Monte Carlo, linear programming; wargame data architecture; analytical decision support products
TM-40H AI Engineer	AI/ML specialists	AIP Logic authoring; Agent Studio; LLM integration (RAG, ontology grounding); AI safety gates; Python transforms for AI consumption; production deployment and monitoring
TM-40I ML Engineer	MLEs, data scientists	Code Workspace model dev; model versioning and experiment tracking; production deployment and Ontology integration; MLOps patterns; responsible AI documentation
TM-40J Program Manager	PMs, G8/S8	Agile PM (Scrum/Kanban) for data/AI; user stories for data products; ML/AI lifecycle management; sprint tracking dashboards; change management and user adoption
TM-40K Knowledge Manager	KMOs, 37F	AAR capture systems; lessons-learned ingestion pipelines; AIP Logic for knowledge summarization; semantic search systems; doctrine/SOP version control; expertise registries
TM-40L Software Engineer	SWEs	OSDK (TypeScript/Python); Foundry Platform SDK; TypeScript Functions on Objects; Action validators; Slate applications; USAREUR-AF code review and deployment standards

NOTE — TM-40J/TM-40K Prereq: TM-30 is required (not recommended) for both TM-40J and TM-40K. Route all PMs and KMOs through TM-30 before enrolling in TM-40J or TM-40K regardless of whether their teams write code. Consult the ENROLLMENT_SOP for waiver procedures.

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TRAINING MANAGEMENT

Administrative Publications

Resource	File	Audience
Mission Training Plan (MTP)	training_management/MTP_MSS.md	Training managers, S3/G6, commanders
Program of Instruction (POI)	training_management/POI_MSS.md	Training managers, curriculum designers
Course Administrative Data (CAD)	training_management/CAD_MSS.md	Training administrators
Training and Evaluation Outline (TEO)	training_management/TEO_MSS.md	Instructors, evaluators
Annual Training Schedule	training_management/ANNUAL_TRAINING_SCHEDULE.md	Training managers, S3
Enrollment SOP	training_management/ENROLLMENT_SOP.md	Training coordinators
Completion Certificate Template	training_management/COMPLETION_CERTIFICATE.md	Training administrators
Faculty Development Plan	training_management/FACULTY_DEVELOPMENT_PLAN.md	Instructor cadre, S3
Training Policy Letter	training_management/POLICY_LETTER.md	Commanders, training managers
Curriculum Maintenance SOP	training_management/CURRICULUM_MAINTENANCE_SOP.md	Curriculum owner, SMEs, instructors
AAR / Feedback Template	training_management/AAR_TEMPLATE.md	Instructors, evaluators

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Lesson Plans

Resource	File
Lesson Plan Template	training_management/lesson_plans/LP_TEMPLATE.md
TM-20 Lesson Plan Outlines	training_management/lesson_plans/TM20_LESSON_PLAN_OUTLINES.md
TM-30 Lesson Plan Outlines	training_management/lesson_plans/TM30_LESSON_PLAN_OUTLINES.md
TM-40 Specialist Lesson Plan Outlines	training_management/lesson_plans/TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md

Course Syllabi

Distribute to each trainee at course start.

Syllabus	File	Audience
TM-10	syllabi/SYLLABUS_TM10.md	All personnel
TM-20	syllabi/SYLLABUS_TM20.md	All staff
TM-30	syllabi/SYLLABUS_TM30.md	Data-adjacent specialists
TM-40A	syllabi/SYLLABUS_TM40A.md	G2/S2 staff, targeting officers
TM-40B	syllabi/SYLLABUS_TM40B.md	FSE, fire support officers
TM-40C	syllabi/SYLLABUS_TM40C.md	G3/S3 maneuver staff
TM-40D	syllabi/SYLLABUS_TM40D.md	G4/S4, logistics staff
TM-40E	syllabi/SYLLABUS_TM40E.md	Force protection, CBRN, PM staff
TM-40F	syllabi/SYLLABUS_TM40F.md	G3/S3 staff, battle captains, CDRs
TM-40G	syllabi/SYLLABUS_TM40G.md	ORSA analysts
TM-40H	syllabi/SYLLABUS_TM40H.md	AI/ML specialists
TM-40I	syllabi/SYLLABUS_TM40I.md	MLEs
TM-40J	syllabi/SYLLABUS_TM40J.md	PMs, G8/S8
TM-40K	syllabi/SYLLABUS_TM40K.md	KMOs, 37F
TM-40L	syllabi/SYLLABUS_TM40L.md	SWEs

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NOTE

TM-50 series syllabi not yet published. TM-50 instruction is managed through the MTP. Contact training management for scheduling.

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SUPPORTING REFERENCE MATERIAL

Resource	File	Notes
MSS Quick Start	QUICK_START.md	30-minute onboarding
Naming & Governance Standards	standards/NAMING_AND_GOVERNANCE_STANDARDS.md	USAREUR-AF MSS naming conventions and governance checklists
MSS Training Hub (web)	mss_info_app/index.html	Standalone HTML training hub — open in browser
Practical Exercises	exercises/	Hands-on packages for TM-10 through TM-40L
Enterprise Data Compass	quick_reference/cda_reference/ENTERPRISE_DATA_COMPASS.md	Authoritative reference — data architecture, ontology, semantic governance
EA vs DA Reference	quick_reference/cda_reference/EA_VS_DA.md	Enterprise Architecture vs Data Architecture — definitions and distinctions
Plan for Success	quick_reference/cda_reference/PLAN_FOR_SUCCESS.md	Technology radar + 5-phase program roadmap with approval chains
Lessons Learned Reference	quick_reference/cda_reference/LESSONS_LEARNED.md	AAR format + LL-001 AVT25 assessment tools case study
CDA Slide Library (course portal)	source_material/course_portal/	29 PDF decks — Intro To Data, Data 101, Data 201 — prereq reading for TM tracks
CDA Interactive Apps	source_material/cda_apps/	Browser-based training tools (EA vs DA, Enterprise Data Compass, LL, Plan for Success)
CDA Final Exam	exercises/exams/EXAM_CDA_FINAL.md	Comprehensive 300-pt exam covering all three CDA slide tracks
Foundry Python Patterns	data_skills/13_foundry_patterns/	Local shim for offline development and testing
Full data science reference library	data_skills/	15 modules: Python → ML → ETL → deployment
SITREP Tracker (applied project)	sitrep_tracker/	Click + Rich + SQLite reference implementation

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QUICK-REFERENCE TRAINING PATHS

New Soldier joining the data team

1. If O-5+: read Data Literacy for Senior Leaders
2. Skim Data Literacy Technical Reference Ch 1–3 and Ch 8
3. Complete TM-10
4. Request MSS access; practice navigation tasks
5. If building: complete TM-20
6. Complete TM-30
7. WFF functional staff: proceed to TM-40A–F per function (after TM-30)
8. Technical specialists: proceed to TM-40G–L per role (after TM-30)

S6 shop standing up MSS capability

1. Commander reads Data Literacy for Senior Leaders
2. All staff complete TM-10
3. Designated builders complete TM-20
4. Designated data leads complete TM-30
5. Reference Glossary throughout
6. Technical specialists proceed to TM-40G–L per role
7. WFF functional staff proceed to TM-40A–F per function

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TM-40 Track Selection by MOS/Role

Role / MOS	TM-40 Track
All WFF staff	TM-40A–F per assigned warfighting function
17A/17C — Cyber officer/NCO	TM-40L (SWE) or TM-40H (AI Eng)
25D — IT specialist	TM-40L (SWE)
25U — Signal support	TM-40L (SWE) or TM-40J (PM)
FA49 — Operations Research	TM-40G (ORSA)
37F — Psychological Operations	TM-40K (KM)
G2/S2 analyst	TM-40G (ORSA) or TM-40K (KM)
G6/S6 data officer	TM-40L (SWE) or TM-40H (AI Eng)
G8/S8 resource manager	TM-40J (PM)
G9/Civil Affairs	TM-40J (PM) or TM-40K (KM)
Data scientist (GS/contractor)	TM-40G (ORSA) or TM-40I (MLE)
AI/ML engineer (GS/contractor)	TM-40H (AI Eng) or TM-40I (MLE)
KMO / Knowledge Officer	TM-40K (KM)
PM / Program Manager	TM-40J (PM)

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